

$$1. F = (\overline{A'(A'D)}) (A'+BC) = (A+D)(A'+BC)$$

$$G = (A'+BC)(A+D')$$

2.

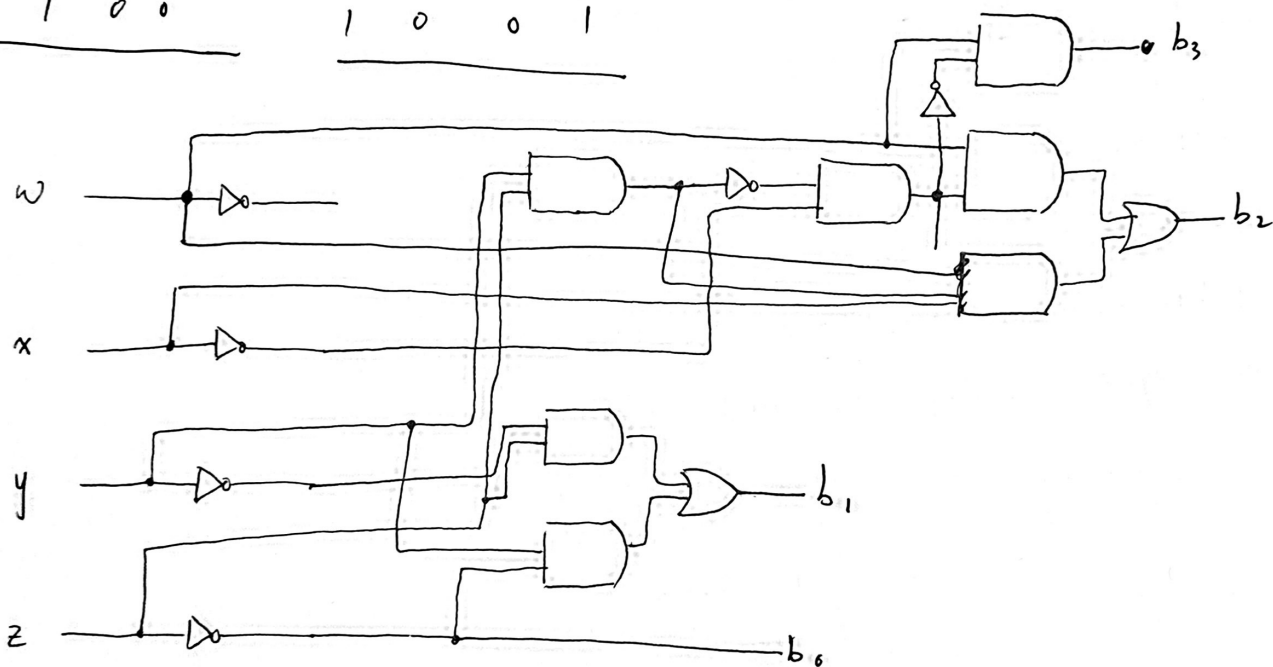
w	x	y	z	b ₃	b ₂	b ₁	b ₀
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	0
0	1	1	0	0	0	1	1
0	1	1	1	0	1	0	0
0	0	0	0	0	1	0	1
1	0	0	1	0	1	1	0
1	0	1	0	0	1	1	1
1	0	1	1	1	0	0	0
1	1	0	0	1	0	0	1

$$b_3 = \cancel{wx} + w = w(x+y+z)$$

$$b_2 = w'xyz + wx'(y'+z')$$

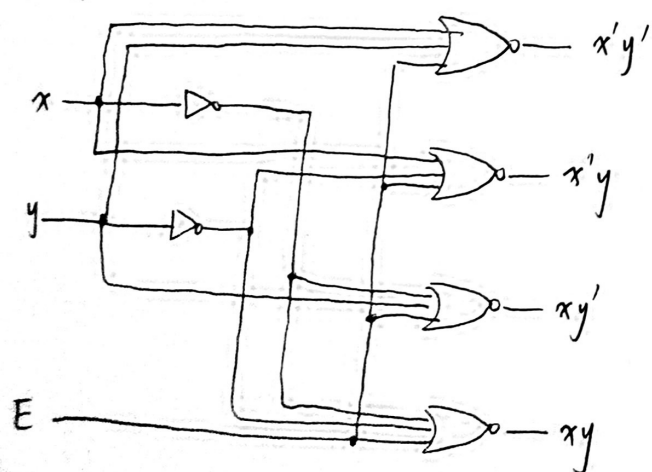
$$b_1 = y'z + yz'$$

$$b_0 = z'$$

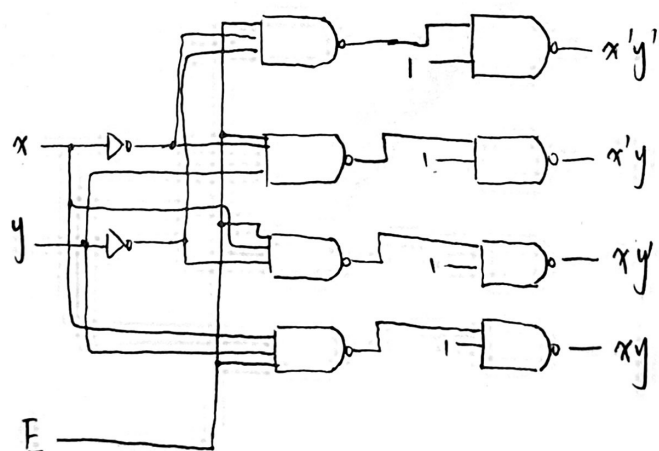


3.

或非



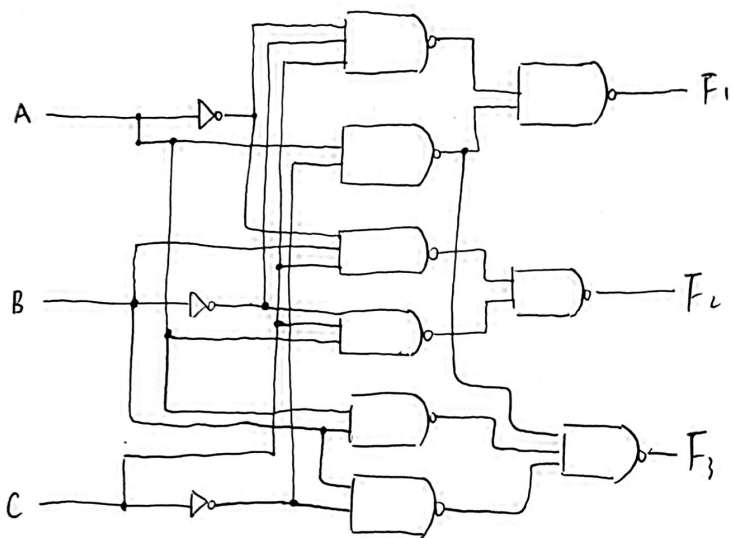
与非



$$4. F_1 = A'B'C + AB'C' + ABC' = A'B'C + AC'$$

$$F_2 = ABC + A'B'C = (A'B + AB')C$$

$$F_3 = BC' + AB + AC = A(B+C') + BC'$$

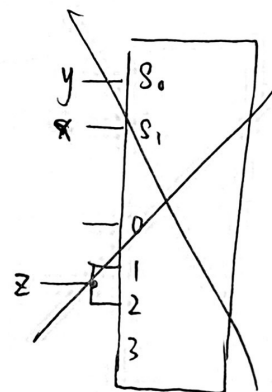
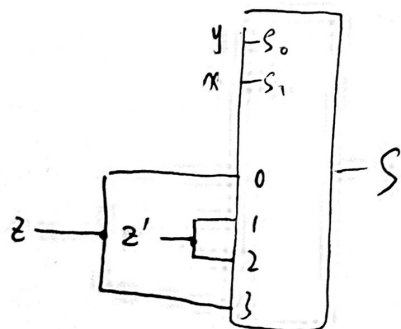
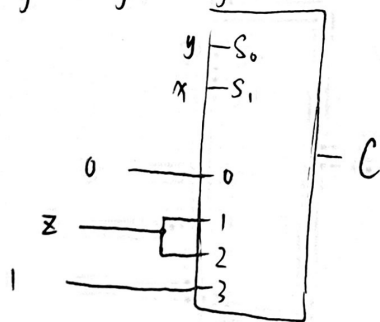
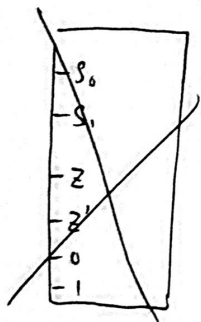


$$5. C = xy + yz + zx, S = \overline{(x=y)z} + \overline{(x+y)z}$$

$$S = \overline{(x-y)z} + \overline{(x+y)[(z=x) + (z=y)]}$$

$$S = \overline{xyz'} + \overline{xy'z} + \overline{x'y'z} + \overline{xyz} = \overline{xy} + \overline{(xy' + x'y)z}$$

$$C = xy + (x+y)z, S = x'y'z + x'y'z' + xy'z' + xy'z$$



6.

	00	01	11	10
00		1		1
01		1	1	
11		1	1	
10	1		1	1

$$\begin{aligned}
 F &= BD + A'C'D + ACD + B'CD' \\
 &\quad + AB'C + AB'D' \\
 &= A'B'(C \oplus D) + A'BD + AB'(C + D') + ABD
 \end{aligned}$$

